

ANNEX B
TERMS OF REFERENCE (ToR) FOR BUSINESS APPLICATIONS DEVELOPMENT
LOT 1C - WEB APPLICATIONS AND OPEN SOURCE RELATED

PART I	
Purpose of Assignment	Design, development, and deployment of the Integrated Campaign Registry (ICR)
Location of Assignment	Remote with in-country visits to 2 countries (TBD)
Duration of contract	17 months
Reporting to:	Sean Blaschke

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1. Background

Public health campaigns remain a critical strategy for preventing and controlling a range of infectious diseases, including vaccine-preventable diseases (VPDs) and neglected tropical diseases (NTDs). However, the delivery of these campaigns has often been verticalized, with each campaign using its own systems, tools, and processes for planning, implementation, and reporting. This fragmentation has led to duplication of effort, inefficiencies in resource use, and missed opportunities for integration. A key challenge is the lack of standardized, interoperable data systems that enable governments and partners to coordinate across different campaigns. Communities are frequently remapped and re-registered for multiple campaigns, while valuable data generated by one program (e.g., polio immunization) is rarely available to inform others (e.g., NTD mass drug administration). This undermines the potential for improved efficiency, equity, and accountability.

UNICEF, as a global leader in campaign digitalization and health systems strengthening, seeks to address this gap by supporting the development of an **Integrated Campaign Registry (ICR)**. **The ICR will provide a digital platform for integrating and reusing campaign data and metadata across programs, ensuring alignment with national health information systems and international reporting standards. By leveraging open standards such as Health Level Seven – Fast Healthcare Interoperability Resources (HL7 FHIR) the ICR will enable greater interoperability, facilitate more effective campaign planning, and improve reporting for governments and partners.** For a technical overview of the ICR, see [Annex 1 – Integrated Campaign Registry](#).

2. Purpose, Objectives and Scope of Work

Purpose and Objectives

The overall **purpose** of this assignment is to support UNICEF in the design, development, and deployment of the Integrated Campaign Registry (ICR). The ICR will serve as a foundational digital public good to integrate and repurpose data and metadata from existing campaign tools, strengthen linkages with routine health information systems, and ensure standardized, interoperable reporting. The ICR will be built in compliance with HL7 FHIR standards to ensure global interoperability and long-term sustainability.

Specific **objectives** include:

- Design and develop a scalable, interoperable ICR that adheres to HL7 FHIR standards and aligns with national and global reporting requirements,
- Facilitate integration and repurposing of data and metadata from existing campaign tools (e.g., immunization, NTD mass drug administration, vitamin A supplementation), avoiding duplication and enabling data reuse across programs,
- Strengthen linkages between community-level reporting and national health management information systems (HMIS), ensuring timely and accurate data flow and improved reporting,
- Enhance capacity of governments and partners in 2 pilot countries to use the ICR for planning, monitoring, and reporting of public health campaigns, supporting long-term institutionalization and sustainability.

Scope of Work (SoW)

The assignment will support UNICEF in the design, development, piloting, and institutionalization of the ICR, a digital platform and standards framework to improve the integration and reuse of metadata and data across public health campaigns, in alignment with UNICEF, WHO, and national priorities. The work will be organized under five main activity areas:

Phase 1 – Development of HL7 FHIR Implementation Guide (IG)

- Review existing campaign data from 3–4 campaigns (UNICEF will help facilitate access) to inform the IG design for the integrated campaign registry.
- Draft FHIR ICR IG and circulate to stakeholders (MoH, UNICEF, WHO, others) for structured feedback.
- Define which components must be reported in real-time vs. reconciled at campaign close.
- Finalize v1 draft IG for use in the initial ICR pilot.

Phase 2 – Development and Deployment of the ICR Platform

- Develop the initial ICR solution and deploy for first in-country testing/piloting.
- In first country, make ICR solution available to enhance the data systems used in two campaigns identified by UNICEF Country Office. Support integration of data using the ICR with two campaigns in the pilot country.
- Update the IG and ICR system based on feedback from the initial pilot.
- Expand to a second country, supporting integrated data management of two additional campaigns identified by UNICEF Country Office.
- Package the ICR solution for replication in other countries, including:
 - Documentation on usage, management, and extension.
 - Design based on open source / access licensed software to ensure sustainability.

Phase 3 – Capacity Building and Training

- In each implementing country, support UNICEF to train MoH and partners on ICR use, health information exchange, and integration into national reporting workflows.
- Provide online job aids and SOPs to enable integration of data across current and future campaigns.
- Work with UNICEF to ensure ICR data can feed into country level microplanning exercises (UNICEF will be responsible for leading the microplanning). Specific tools used for microplanning will be country-dependent but will benefit from harmonized, enriched ICR data.

Phase 4 – Alignment with Global and National Reporting Systems

- Ensure UNICEF-supported MDA campaigns report using JAP-aligned data formats.
- Support UNICEF DAPM team to help access ICR data to support their real-time monitoring work
- Support export of data aligned with JAP forms.

Phase 5 – Systems Integration

- Develop analytics add-on that will allow automatic syncing of ICR data to a data warehouse to make ICR data (structured FHIR data) more accessible for external analytical uses.
- Develop data connectors to the ICR that will allow the importing and exporting of data with DHIS2 and one additional UNICEF designated platform, i.e., Open Data Kit (ODK).
- Establish two-way integration with WHO Geospatial Microplanner to improve data use in microplanning and to push data updates back into the ICR.

Phase 6 – Sustainability and Continuity

- Embed ICR reporting processes into national Ministry of Health reporting cycles.

- Strengthen local capacity for ongoing system maintenance, data governance, and future extensions beyond the grant period.
- Provide a replication toolkit to facilitate expansion of the ICR to additional countries.

Expected Outputs

The following are the main expected outputs from the selected Vendor:

- A standardized HL7 FHIR Implementation Guide for campaign data is developed and validated for use in pilot countries
- An operational ICR platform is developed, piloted in two countries, and packaged for replication
- Ministries of Health and implementing partners in two countries have the capacity to use and maintain the ICR
- ICR is interoperable with global and national reporting requirements (JAP) and platforms, i.e., DHIS2
- ICR is institutionalized within national reporting processes and supported for replication

ICR is sustainably embedded within national systems, with strengthened government capacity for ongoing maintenance, data governance, and scale-up beyond pilot countries.

3. Key Deliverables and Timelines

Phase	Overall Phase Output	Phase Activities	Expected deliverables	Timeline
1 - Development of HL7 FHIR Implementation Guide (IG)	A standardized HL7 FHIR Implementation Guide for campaign data is developed and validated for use in pilot countries	▪ Review existing campaign data from 3–4 campaigns (UNICEF will help facilitate access) to inform the IG design for the integrated campaign registry. ▪ Draft FHIR ICR IG and circulate to stakeholders (MoH, UNICEF, WHO, others) for structured feedback. ▪ Define which components must be reported in real-time vs. reconciled at campaign close. ▪ Finalize v1 draft IG for use in the initial ICR pilot.	a. Project workplan developed and validated with UNICEF, outlining approach to IG development and stakeholder engagement. b. Draft HL7 FHIR IG for campaign data. c. Documented definition of data elements and reporting requirements, including specification of components to be reported in real-time versus reconciled at campaign close. d. Revised HL7 FHIR IG delivered and endorsed for pilot.	Months 1-2
2 - Development and Deployment of the ICR Platform	An operational ICR platform is developed, piloted in two countries, and packaged for replication	▪ Develop the initial ICR solution and deploy for first in-country testing/piloting. ▪ In first country, make ICR solution available to enhance the data systems used in two campaigns identified by UNICEF Country Office.	a. Prototype ICR system built and tested in first pilot country. b. Documented workflows of integration of two campaigns in pilot country. c. Updated IG and system (v1.1) incorporating pilot feedback.	Months 3-6

		Support integration of data using the ICR with two campaigns in the pilot country. ▪ Update the IG and ICR system based on feedback from the initial pilot. ▪ Expand to a second country, supporting integrated data management of two additional campaigns identified by UNICEF Country Office. ▪ Package the ICR solution for replication in other countries, including: Documentation on usage, management, and extension, and Design based on open source / access licensed software to ensure sustainability.	d. Second-country deployment with at least two campaigns integrated. e. Packaged ICR solution with: ○ User manual and admin guide. ○ Technical documentation for replication. ○ Open-source or low-cost licensing framework.	
3 - Capacity Building and Training	Ministries of Health and implementing partners in 2 countries have the capacity to use and maintain the ICR (NOTE: Pilot countries will be identified after the country engagement phase during the project inception phase.)	▪ In each implementing country, support UNICEF to train MoH and partners on ICR use, health information exchange, and integration into national reporting workflows. ▪ Provide online job aids and SOPs to enable integration of data across current and future campaigns. ▪ Work with UNICEF to ensure ICR data can feed into country level microplanning exercises (UNICEF will be responsible for leading the microplanning). Specific tools used for microplanning will be country-dependent but will benefit from harmonized, enriched ICR data.	a. Training and/or Training-of-trainers (ToT) sessions conducted in each pilot country for MoH and implementing partners. b. Job aids and SOPs developed to support integration of ICR data across current and future campaigns. c. Documented approach and implementation of integration of ICR data into national reporting and microplanning workflows.	Months 7-12
4 - Alignment with Global and National Reporting Systems	ICR is interoperable with global and national reporting requirements and platforms	▪ Ensure UNICEF-supported MDA campaigns report using JAP-aligned data formats. ▪ Support UNICEF DAPM team to help access ICR data to support their real-time monitoring work. ▪ Support export of data aligned with JAP forms.	a. JAP-aligned data outputs and export functionality developed and validated for UNICEF-supported MDA campaigns. b. Access workflows established and documented to enable use of ICR data by UNICEF DAPM for real-time monitoring.	Months 13-17

			c. Reporting outputs and data exports aligned with JAP forms and validated against WHO and national reporting requirements.	
5 - Systems Integration	ICR is institutionalized within national reporting processes and supported for replication	- Develop analytics add-on that will allow automatic syncing of ICR data to a data warehouse to make ICR data (structured FHIR data) more accessible for external analytical uses. - Develop data connectors to the ICR that will allow the importing and exporting of data with DHIS2 and one additional UNICEF designated platform, i.e., Open Data Kit (ODK). - Establish two-way integration with WHO Geospatial Microplanner to improve data use in microplanning and to push data updates back into the ICR.	a. Analytics add-on developed to enable automatic synchronization of ICR structured FHIR data to a data warehouse for external analytical use. b. Data connectors developed and implemented to support import and export of data between the ICR, DHIS2, and one additional designated platform. a. Two-way integration established and tested between the ICR and WHO Geospatial Microplanner.	Months 13-17
6 - Sustainability and Continuity	ICR is sustainably embedded within national systems, with strengthened government capacity for ongoing maintenance, data governance, and scale-up beyond pilot countries	- Embed ICR reporting processes into national Ministry of Health reporting cycles. - Strengthen local capacity for ongoing system maintenance, data governance, and future extensions beyond the grant period. - Provide a replication toolkit to facilitate expansion of the ICR to additional countries.	a. ICR reporting processes embedded and documented within national MoH reporting cycles. b. Capacity strengthening implemented for ongoing system maintenance, data governance, and future system extensions (documentation, source code, maintenance guide). c. Replication toolkit (guidelines, templates, documentation) developed for expansion to additional countries.	Months 13-17

4. Proposal Structure

Technical and Financial Proposal Content

The proposal must cover the following sections (additional sections are welcome):

1. Detailed understanding of UNICEF requirements.

2. Expertise of Company detailing general and specific experience with similar assignments in size, scope, and complexities in the past five years. Proposers are requested to back up their submissions by providing evidence in the form of job completion certificate, contracts and/or references. This should include:
 - Published, inspectable FHIR IG in public registry or GitHub (disqualifying if absent): the Vendor must provide a live link to a published FHIR Implementation Guide — not a description of one, a slide deck, or a reference to API integrations. Reviewers will inspect the IG directly: profiles, narrative, and structure. Links to GitHub repositories containing FSH/SUSHI source files are acceptable provided the IG has been compiled and is browsable. Submissions that do not include a working link to an inspectable IG will be disqualified.
 - Working FSH/SUSHI profiles submitted as part of the application (disqualifying if absent): the Vendor must submit sample profiles as part of the proposal (minimum: one StructureDefinition, one ValueSet, one Extension). These will be validated against FHIR R4. Profiles that do not compile or that reference undefined base resources will be treated as non-compliant. The intent is to distinguish Vendors who write FHIR from Vendors who consume it.
 - Named FHIR IG author on the proposed team with verifiable CV (disqualifying if absent): the proposal must name the specific individual who will author the ICR FHIR IG, provide their CV, and list prior IGs they have authored that are publicly inspectable. "To be recruited" is not acceptable. The named individual must be confirmed on the contract at award. UNICEF reserves the right to verify authorship claims against public IG registries.
 - Experience developing WHO SMART Guideline-based L3 applications in FHIR: the Vendor should describe specific L3 implementations, referencing the WHO SMART Guidelines methodology. Generic statements about FHIR experience will not be scored. Evidence of engagement with the HL7 community (publications, co-authorship of IGs, participation in connectathons) is an asset.
 - Demonstrated experience supporting digital campaign implementation (e.g. immunization, NTD MDA, IRS, etc.): the Vendor should describe specific campaign implementations with named countries and partners. Understanding of campaign registry data flows — including coverage tracking, campaign deduplication, Gavi/COVAX reporting, and DHIS2/OpenSRP ecosystems — is required.
 - Demonstrated experience integrating and deploying geospatial microplanning campaign tools: the Vendor should describe specific geospatial integration work, ideally with WHO Geospatial Microplanner or equivalent tools used in LMICs. Experience with GIS-based household enumeration, settlement-level coverage mapping, and data push/pull between campaign registries and microplanning layers is relevant.
 - Demonstrated experience having developed and contributed to a recognized Digital Public Good (DPG): the Vendor should reference the specific DPG, their role (contributor, maintainer, lead developer), and provide a link to the DPG registry entry. Contributions limited to documentation or minor bug fixes will be weighted lower than active co-development or governance roles.
 - Demonstrated experience with data integrations on production systems (DHIS2, OpenSRP, ODK or similar): the Vendor should describe production integrations — not sandboxes or prototypes. Relevant experience includes terminology management (CVX codes, ConceptMaps between code systems, binding strengths for vaccine products and target populations), bidirectional sync, and management of data flows in low-connectivity environments. Named country deployments preferred.
 - Demonstrated experience leading complex initiatives with national and local governments across multiple regions and countries: the Vendor should provide evidence of multi-country programme management experience, including engagement with Ministries of Health, coordination with UN

agencies, and management of culturally and linguistically diverse stakeholder groups. French language capacity (spoken and written, able to lead capacity-building workshops) is required.

3. Detailed Methodology/Approach demonstrating how the Vendor meets or exceeds UNICEF requirements for this assignment as detailed above.
4. Completed Tables A, B and C of Appendix 2 – UNICEF Information Security and Platform & Infrastructure Requirements.
5. Proposed work plan and timeline for the deliverables mentioned above and a clear process to coordinate with UNICEF technical team.
6. A proposal of the working method between the Vendor and UNICEF that efficiently meets the objectives defined.
7. Details of the Proposed Team for the assignment including the following information: related technical skills, experience in working on similar project and assignment, list of similar projects they worked on and their roles on the project. If the bidder is expecting to hire for this role, please indicate so clearly.
8. Any dependencies and assumptions, where applicable.
9. Detailed budget proposal broken down as per instructions in Annex C – Pricing Tables (the Financial Proposal).
10. In line with optional Maintenance and Support (Section 7, Period Performance), the Vendor is expected to submit a comprehensive Service Level Agreement (SLA) for maintenance and support services. The proposed SLA shall, at a minimum, include:
 - A description of the support model (e.g., helpdesk structure, ticketing system, and escalation procedures);
 - Clearly defined response and resolution timelines based on severity levels;
 - The approach to issue tracking, reporting, and resolution management; and
 - The availability of support services, including time zone coverage and hours of operation.
 - Outline post-warranty support arrangements, including considerations for long-term sustainability and any proposed approach for a potential transition to government ownership, where applicable.

Partnerships, Consortium and Subcontracting Arrangements

Bidders may propose to fulfil this engagement as part of a **partnership, consortium or through subcontracting arrangements**, provided that **the LTAS holder is designated as the Prime/Lead Contractor**.

The Prime/Lead Contractor shall retain full accountability and contractual responsibility to UNICEF for performance of the entire assignment including all activities carried out by consortium/partnership/subcontracting members.

Where a **consortium/partnership** is proposed, **the proposal submission must:**

1. Clearly define the Prime/Lead Contractor (LTAs holder) and all participating partners;
2. Define roles and responsibilities of each entity;
3. Present a clear governance and coordination structure for the engagement;
4. Include signed letters of commitment from all consortium partners confirming their participation;
5. Disclose all partners at the time of proposal submission. Any changes or substitution of partners after contract award shall require prior written approval from UNICEF.

Where **subcontracting** is proposed, bidders must:

1. Clearly identify all proposed subcontractor(s);

2. Specify the scope of work to be subcontracted;
3. Indicate the percentage of the total contract value to be subcontracted;
4. Ensure all subcontractors comply with UNICEF's applicable policies and requirements.

UNICEF reserves the right to review, approve or reject any proposed consortium members or subcontractors.

Bidders are encouraged to demonstrate that any proposed consortium/partnership or subcontracting arrangement complements and strengthens the overall technical and operational capacity required under this ToR and does not replace the core competencies expected from the Prime/Lead Contractor.

5. Proposal Evaluation Methodology and Scoring Structure

The evaluation will be carried out by UNICEF in accordance with UNICEF's regulations, rules, and practices and all determinations are made in UNICEF's sole discretion.

The assessment of proposals has the following breakdown:

- Technical Proposal - 70%
- Financial Proposal (Filled in Annex C - Pricing Tables) - 30%

The overall combined score shall be computed as:

Combined score = (Technical Score × 70%) + (Financial Proposal Score × 30%)

Disqualifying criteria

The following criteria are pass/fail. Proposals that do not satisfy all three will be disqualified from further evaluation, regardless of overall technical quality:

#	Criterion	Evidence required
1	Published, inspectable FHIR IG in public registry or GitHub	The Vendor must provide a live link to a published and inspectable HL7 FHIR IG in a public registry or GitHub repository. The IG must be accessible and reviewable (profiles, narrative, and structure). Descriptions, slide decks, or references to APIs are not sufficient.
2	Working FSH/SUSHI profiles submitted as part of the application	The Vendor must submit sample FHIR artefacts as part of the proposal, including at minimum one StructureDefinition, one ValueSet, and one Extension. These must be compliant with FHIR R4 and technically valid.
3	Named FHIR IG author on proposed team with verifiable CV	The proposal must identify a named FHIR IG author who will be responsible for IG development under this assignment. The individual's CV must be included, along with references to previously authored, publicly accessible IGs. "To be recruited" profiles will not be accepted.

Technical Evaluation Criteria

Technical proposals will be assessed based on criteria presented below. Bidders may receive maximum technical score of 100 points.

1. **Technical Approach and System Design (40 points)** - Assessment of the overall technical quality and feasibility of the proposed approach, including:
 - Clarity and robustness of the methodology for developing the ICR IG, including validation and publication processes
 - Approach to enhancing existing country systems and registries rather than creating parallel platforms
 - Strategy for interoperability with national HMIS (e.g., DHIS2) and other campaign-related digital tools
 - Approach to data standardization, reuse, and integration across campaigns and programmes
 - Consideration of scalability, sustainability, and alignment with government digital health architectures
2. **Campaign and Immunization Domain Expertise (30 points)** - Assessment of the Vendor's demonstrated understanding of campaign digitization, including:
 - Familiarity with immunization and campaign data standards, including WHO SMART Guidelines where applicable
 - Understanding of campaign registry data flows, including coverage tracking, deduplication, reporting requirements (e.g., Gavi, COVAX), and integration with DHIS2 and related platforms
 - Experience managing health terminology standards (e.g., CVX codes, ConceptMaps, value sets)
 - Demonstrated experience supporting digital implementation of campaigns (e.g., immunization, NTD MDA, malaria IRS, or similar large-scale public health interventions)
3. **Country Implementation, Partnerships, and Capacity Building (30 points)** - Assessment of the Vendor's ability to deliver in real-world country contexts and support sustainable institutionalization, including:
 - Demonstrated experience working with Ministries of Health and supporting national or subnational digital health systems
 - Ability to deliver in LMIC contexts, including presence or strong operational experience in relevant regions
 - Experience coordinating with multiple stakeholders, including governments, UN agencies, donors, and implementing partners
 - Quality and feasibility of the proposed capacity-building and knowledge transfer approach, including training, documentation, and handover to government counterparts
 - Demonstrated ability to integrate digital solutions within existing government systems and workflows, avoiding the creation of parallel systems

Financial Evaluation Criteria

Bidders are required to submit their Commercial Proposals in accordance with the instructions and as per the format provided in Annex C ("Pricing Tables").

The methodology for allocating points among Commercial proposals is as follows:

$$\text{Score for price proposal X} = \frac{30 \text{ (maximum points)} * \text{Price of lowest priced proposal}}{\text{Price of proposal X}}$$

6. Administrative Arrangement

UNICEF Responsibilities: UNICEF shall have no obligation to provide any assistance to the Vendor in performing the services other than as expressly set forth herein:

- Provide information and feedback in a timely manner and as long as Vendor's requests for information and feedback are also made to UNICEF within reasonable time.
- Provide the Vendor with reasonable and timely access to UNICEF personnel as may be required to perform the services.
- Provide Vendor's personnel with access to UNICEF internal systems where applicable, physical access to the necessary work area and equipment and any required identification badges and pass codes before work commences.
- A single point of contact responsible for representing UNICEF and assisting with the decision -making process regarding the engagement.

Chosen Bidder Responsibilities: Chosen bidder will be expected to:

- Review campaign data structures and develop a standardized HL7 FHIR Implementation Guide.
- Design, build, and deploy the Integrated Campaign Registry (ICR) in two pilot countries.
- Provide technical documentation, user manuals, and replication materials for the ICR.
- Conduct online training and capacity-building activities for Ministries of Health and implementing partners.
- Ensure interoperability of the ICR with national health information systems and WHO reporting requirements.
- Support institutionalization of the ICR within national reporting processes and provide sustainability and handover packages.

Shared Responsibilities:

- Testing and validation: UNICEF will provide access to relevant country-level test data and facilitate coordination with MoH counterparts; chosen bidder will design and execute test plans, document results, and lead resolution of identified issues.
- Deployment: UNICEF will coordinate in-country access, stakeholder availability, and MoH readiness; chosen bidder will manage technical deployment, configuration, and go-live support.
- Feedback and iteration: UNICEF will consolidate and communicate stakeholder feedback within agreed timelines; chosen bidder will incorporate feedback into subsequent development cycles per the agreed work plan.
- Acceptance: UNICEF will review and formally accept deliverables against agreed criteria; chosen bidder will provide supporting evidence of compliance as required.

7. Project Particulars

Period of Performance

The initial contract engagement is for 17 months with estimated start date on 1 May 2026 and ends on 30 Sept 2027. Services under contract to start immediately upon contract signature.

UNICEF reserves the right to extend ("OPTIONAL") the Contract to accommodate for maintenance and support services, for an additional period of up to four (4) years, 12 months each, subject to the satisfactory performance of

the Vendor. Any such extension shall be under the same terms and conditions, including pricing, as set out in Annex C - Pricing Tables, and the Vendor shall acknowledge that such extension is **optional and not guaranteed**.

In the event that additional or new development requirements arise that fall outside the scope of this ToR, such services shall be managed through a formal Change Order process, with the scope, pricing, and timeline to be mutually agreed between the Parties through Contract Amendment.

UNICEF further reserves the right to cancel the contracted services, in whole or in part, at any time and without cost, in accordance with its contractual rights.

Location

The services are expected to be performed remotely (offsite from the vendor's premises). However, limited in-country travel may be required. It is anticipated that the selected vendor **may ("OPTIONAL") undertake one (1) trip per pilot country (two countries in total, Sub-Saharan Africa)**, with a maximum duration of ten (10) days per trip, to support testing of the initial ICR prototype.

Travel-related costs shall be proposed separately in Annex C – Pricing Tables, in accordance with the provisions on Reimbursable Travel set out in Article 6.1 of LTAS Annex D – Fee Schedule.

All travel and associated costs shall be subject to prior written approval by UNICEF, and no travel expenses shall be incurred or reimbursed without such approval.

Engagement Type

The engagement shall be structured as a Time and Materials (T&M) contract, with clearly defined ceilings to ensure cost control and transparency. The Vendor shall propose, in Annex C – Pricing Tables, the maximum not-to-exceed (NTE) amount per phase and per deliverable, which shall constitute the agreed financial cap that both Parties will adhere to throughout the Contract.

Payment shall be made based on the actual time and effort expended, subject to the completion and acceptance of deliverables (in full) and within the agreed NTE amounts per phase/deliverable. Under no circumstances shall payments exceed the maximum amounts specified in Annex C - Pricing Tables without a formally approved amendment or change order.

The Vendor shall ensure that all costs, including professional fees, travel, and any incidental expenses, are appropriately reflected in the pricing structure submitted in Annex C - Pricing Tables.

Please refer to Section 4 – Key Deliverables and other relevant sections of the ToR for the detailed scope, requirements, and timelines associated with each phase and deliverable.

Payment Terms

For Professional Services: (i.e. Building systems development, major development enhancements. re/design and re/build of systems - Phases 1 - 6): The engagement shall be implemented under a Time and Materials (T&M) model, with pre-agreed maximum not-to-exceed (NTE) amounts per Phase and Deliverable as defined in Annex C -

Pricing Tables. Payment shall be made based on the actual time and effort expended **per deliverable**, within the agreed NTE ceilings for each Phase and corresponding deliverables.

Payments will be made per deliverable Net 30 days following:

- Receipt of accurate and complete invoice(s) and satisfactory performance of services rendered, as evidenced by the completion and formal acceptance of the relevant deliverables (in full) by UNICEF.
- Unless otherwise agreed between the Parties, payments will be made per deliverable, based on the accepted deliverables and the actual effort incurred, within the agreed NTE amount for that deliverable.
- The timely performance and completion of tasks/activities shall be assessed against the Project Plan and agreed timelines and must be aligned with the deliverables defined in the ToR.

Under no circumstances shall the total payments exceed the maximum NTE amounts specified in Annex C, unless otherwise agreed through a formal contract amendment or change order.

Recurring Services (i.e. Maintenance and Support Services): If exercised by UNICEF, shall be provided as an **optional component of the engagement**.

Payment for such services shall be made on a quarterly basis in arrears, withing thirty (30) days following:

- Receipt of accurate and complete invoice(s); and
- Confirmation of satisfactory delivery and acceptance of the service by UNICEF.

The total payments for maintenance and support services shall not exceed the annual maximum not-to-exceed (NTE) amount specified in Annex C – Pricing Tables.

The Vendor shall ensure that the proposed annual pricing for maintenance and support remains valid for the entire optional extension period of up to four (4) years, structured as annual periods of twelve (12) months each, under the same terms and conditions, unless otherwise agreed by the Parties through a formal amendment.

Applicable Terms and Conditions

Vendor is reminded that any contract resulting from this TOR will be governed by the terms and conditions of the LTAS for the provision of several categories of Business Application Development Services on both open-source software and proprietary mainstream technologies; valid until 14 June 2026 as agreed between the parties. Therefore, the vendor should avoid in its proposal/response to this TOR any terms and conditions (including pricing) that are inconsistent or in contradiction to the terms and conditions in the LTAS.

8. Others

The Vendor to describe if there is/are any assumptions, Dependencies, Limitations, and Exclusions in this engagement, particularly if additional Roles and/or Responsibilities are to be played by UNICEF during the contract.

8.1 Clarification on Integrations and Responsibilities:

Integrations under this assignment are limited to country-owned systems and globally recognized platforms (e.g., DHIS2, OpenSRP, ODK, and other government-approved tools). No integration with UNICEF internal systems is required. The Vendor will be responsible for:

- Designing and implementing interoperability layers aligned with HL7 FHIR standards;
- Supporting integration with national HMIS and campaign-related systems;
- Developing and documenting data exchange workflows and architecture;
- Ensuring alignment with national digital health strategies and enterprise architecture.

UNICEF will be responsible for:

- Facilitating coordination with Ministries of Health and relevant partners;
- Supporting access to documentation, standards, and existing system architecture where available;
- Providing technical oversight and validation of deliverables.

API availability will depend on country system readiness. UNICEF does not guarantee the availability of APIs from government systems and will not provide APIs from UNICEF systems under this assignment.

9. Qualification Requirements

- Proven experience in designing and implementing digital health solutions in LMICs.
- Demonstrated expertise with HL7 FHIR standards.
- Experience in developing and deploying interoperable systems with national health management information systems (e.g., DHIS2, ESPEN tools).
- Capacity to deliver solutions aligned with open-source and digital public goods principles (i.e., DHIS2, OpenSRP, and others).
- Strong knowledge of data protection and information security standards.
- Ability to provide technical documentation, user manuals, and replication toolkits.
- Experience in training and capacity building on digital health solutions for Ministries of Health and partners.
- Availability of a multidisciplinary team (system architect, developers, interoperability specialist, training expert, project manager).
- Prior work with UN agencies, WHO, Gavi, or Ministries of Health is an asset.
- Demonstrated ability to deliver on time across multiple countries.

10. Appendices

Appendix 1 – Integrated Campaign Registry

What is the Integrated Campaign Registry?

The Integrated Campaign Registry (ICR) is a digital platform that enables public health campaigns — including neglected tropical disease (NTD) mass drug administration, immunization, malaria prevention, and vitamin A supplementation — to share, exchange, and reuse metadata and data in a structured and interoperable way. Traditionally, each campaign has collected data separately, often using different formats and systems. This fragmentation leads to duplication, inefficiencies, and data silos. The ICR acts as a metadata and data backbone, standardizing how campaign data is stored and exchanged. Information from one campaign (e.g., household locations, coverage levels, or commodities distributed) can be repurposed by another, instead of being re-collected. The ICR is designed to ensure alignment with national health management information systems (HMIS) and World Health Organization (WHO) program reporting requirements. In doing so, it contributes not only to better campaign outcomes but also to long-term health system strengthening.

Key Components of the Integrated Campaign Registry

1. FHIR Implementation Guide (IG)
 - A technical “rulebook” on how to store campaign data based on HL7 FHIR standards.
 - Defines how campaign data (locations, households, services delivered, coverage results) should be structured, shared, and reused.
 - Ensures interoperability between platforms like DHIS2, ESPEN tools, or mobile apps (e.g., Open Data Kit, ODK).
2. Core Registry Database
 - Centralized store of campaign records.
 - Receives data from multiple campaign platforms (e.g., polio via ODK, NTDs via DHIS2 Tracker, etc.).
 - Provides APIs for integration with country HMIS.
3. Integration and Data Flow Tools
 - Middleware (e.g., OpenFn / Airbyte connectors) to extract, transform, and load campaign data into the ICR.
 - Export modules for WHO Joint Application Package (JAP) reporting.
 - Sync modules to link ICR with DHIS2 and other Ministry of Health systems.
4. Data Model Elements
 - **Patients and Households:** name, age, gender, household identifiers.
 - **Locations:** settlements, household GPS coordinates, schools, health facilities, health facility catchment areas and other service points.
 - **Services Received:** i.e., vaccinations (by antigen), insecticide treated nets (ITNs), mass drug administration (MDA), indoor residual spraying (IRS), and others.
 - **Campaign Metadata:** dates, target population, implementing partners, coverage levels
5. Analytics Layer
 - Provides structure and data normalization that enables campaign data to be able to flow into a data warehouse for analysis and monitoring.
 - Dashboards and visualizations provide insights for planning and monitoring.
 - Promotes reuse of campaign data across programs (e.g., identifying zero-dose and under-immunized children, mapping underserved areas, etc.).

- Provide data to feed into microplanning workflows to normalize data and enable integration with data warehouses.
 - Dashboards and visualizations for planning, monitoring, and course correction.
 - Reuse of campaign data across programs (e.g., identifying zero-dose and under-immunized children, mapping underserved areas, etc.).
6. Data Enrichment Layer:
- Integration with external tools like WHO Geospatial microplanner to enrich data in campaigns. This can include adding facility catchment areas to health facilities, supervisory zones, population data to administrative areas, risk score assignment to settlement areas, etc.
7. Capacity Building and Sustainability
- Training materials for Ministry of Health staff and campaign managers.
 - Open source/low-cost deployment to minimize licensing barriers.
 - Locally customizable to reduce long-term dependency on external providers.
 - Documentation and replication toolkit to enable adoption in other countries.

Integrated Campaign Registry Workflow

1. Data Collection
 - Campaigns continue using their existing tools (ODK, DHIS2, etc.).
 - Data is formatted via the FHIR IG and passed into the ICR using interoperability modules.
2. Integration
 - ICR serves as a central hub, aligning campaign data across programs.
 - Data is normalized and harmonized (e.g., households, locations) for comparability.
 - Interoperable with Ministry of Health systems and WHO reporting formats.
3. Analysis and Use
 - Integrated data feeds dashboards and data warehouses for real-time insights.
 - Decision-makers can view cross-campaign coverage, overlaps, and gaps.
4. Reporting and Sustainability
 - Campaign results exported in JAP formats for WHO/AFRO/Expanded Special Project for the Elimination of Neglected Tropical Diseases (ESPEN).
 - Workflows embedded into HMIS to ensure continuity beyond donor funding streams.

Appendix 2 – UNICEF Information Security and Platform & Infrastructure Requirements

This section describes UNICEF’s security and platform & infrastructure requirements for systems which store and/or processes internal data and information that, while not sensitive, is limited to select UNICEF business partners.

The security requirements outlined in this document are mandatory (unless indicated as “Optional”) and apply to any internal or external party who is providing a solution, a system or a service to UNICEF which processes, stores or transmits information that meets the classification criteria reflected in this document.

It shall be noted that the requirements outlined below be viewed as an additional layer to complement Vendor’s existing security ecosystem and not a replacement. In cases where a service provider’s controls are more restrictive, the service provider’s controls shall prevail and be formally captured by both parties.

Vendors are required to provide evidence and other documents (or active URL links) to support their claim of compliance.

Table A – Security Requirements

Requirement	Complied? (Yes / No)	Proposer’s Proposal Page/Section Reference for explanation and supporting documents
General Security Requirements <i>Assessment of, or validation of connectivity controls between UNICEF and any third party / service provider are required before any physical or logical connection can be made.</i> <i>Internally hosted systems that connect and access a Cloud system will use UNICEF standard network deployment topology for the following system components:</i> - For frontend systems - In between systems / containers - Databases <i>End point security controls will be implemented similar if not the same to the ones listed for the top 20 CIS controls (https://www.cisecurity.org/controls/cis-controls-list/). These controls shall be kept functional and updated considering the following domains:</i> - Virus protection - Data exfiltration - Unauthorized access to and/or changing of critical system(s)/application files - Zero-day vulnerabilities		
Vendor agrees to the rights of UNICEF to assess the quality and accurateness of outsourced software development and operational maintenance of the system / application; whether it be through security assurance testing or through external security assessment.		
Solution / Service shall be protected from unwanted network traffic by network filtering		

or separating measures that lay outside of the system such as, externally controlled routers and firewalls.		
The system shall have proper end-point protection with the following minimum requirements: • Malicious code protection measures • Host firewall configured utilizing, at a minimum, least privileged access controls (services, user, communication access)		
Validation of Security Controls <i>Impartial security and vulnerability assessment testing shall be performed to determine the effectiveness of security controls for the following system components:</i> • <i>For frontend systems</i> • <i>In between systems / containers</i> • <i>Databases</i> <i>Independent penetration testing agent or teams will perform penetration testing on information systems prior to production release.</i>		
Vendor agrees to the rights of UNICEF to periodically validate the implementation of the security requirements outlined in this document via: • Security Assurance Testing • Vulnerability Testing • Penetration Testing • Audits • On-site checks		
Compliance & Certifications (OPTIONAL) <i>For all systems that are interconnected, the Vendor shall meet IS27001 security controls and/or SOC-2 security controls.</i> <i>These controls can also be referenced from the Cloud Security Alliance Matrix (https://cloudsecurityalliance.org/research/cloud-controls-matrix/).</i>		
Vendor shall carry ISO2700K certification and/or SOC-2 audits as mandatory or other international standards. The Vendor shall provide information about the certification and/or the SOC-2 report and audit findings or equivalent certifications and/or reports.		
Identification, Authentication and Authorization <i>These new groups/ roles created must follow the principle of least privilege requiring that these groups/ roles be given the minimum privileges necessary for their job function.</i>		

By default, all information systems will rely on existing organization wide security groups and roles that exist on UNICEF's Identity Access Management Services. In cases where applications / services permission cannot be matched to existing groups / roles, a request for new application specific groups / roles creation must be submitted for the following system components:

- *Frontend systems.*
- *External partner / other systems/ accounts.*
- *In between systems / containers. This includes the Staging area.*
- *For privileged accounts accessing the backend systems such as the databases.*

All UNICEF staff accounts shall be created on an automated fashion, based on the connection between the IAM System.

The service provider shall follow the principle of least privilege, guaranteeing that users, group, role, and device identifiers will be unique, assigned to each entity (user or process). Each application user role shall have a correspondent database connection according to its privileges.

The service provider shall centrally manage the user account using federated identities and whenever possible integrate their solution with the UNICEF Identity Management System. In case authentication is password based; the password shall forcefully adhere to the common best practice quality requirements and will be forcefully renewed frequently. The allocation of authenticators will be controlled and management through a formal process.

Multi-factor authentication will be used for:

- privileged accounts and
- user access outside of UNICEF trusted network.

All the user and system accounts shall be disabled after a defined period of inactivity, in accordance with organizational standards. All default accounts and or passwords shall be removed or changed. Approvals will be required for creation, deletion, or modification of any account.

Account lockout features will be used for invalid authentication attempts.

Application code shall never contain any credentials.

Availability and Deletion

The system will have a HIGH Level of Availability. Availability requirements and recovery mechanisms shall comply with UNICEF BCM requirements covering critical business processes for the following system components:

- *For frontend systems*
- *In between systems / containers*

- *Databases*

For the deletion of data, a process is required for deletion of any information system data.

Any deletion of confidential / personal data must be done so that it cannot be reconstructed.

Cryptography

Strong cryptographic algorithms shall be used for encrypting data at rest with a minimum standard of Advanced Encryption Standard (AES) 256, or in accordance with existing industry best practice or published UNICEF standard, whichever is more restrictive.

Data on the following systems must follow the above minimum standards while in transit and at rest for the following system components:

- *For frontend systems*
- *In between systems / containers*
- *Databases*

The organization will escrow encryption keys to maintain the availability of information in the event of loss of encryption keys.

Where applicable, key management guidelines from a trusted Cloud service provider will be followed.

The service provider shall use best practice or industry standard secure data exchange protocols and keep them up to date, as per defined UNICEF standards. Outdated and / or compromised protocols shall never be used. In cases where Vendor cryptographic standards exceed published organizational standards, Vendor technical controls shall prevail.

All passwords shall be encrypted with best current practices or strong industry standards cryptographic algorithms and secure keys. The keys will be generated using strong cryptographic algorithms.

Key files must be protected from unauthorized modification using an application that enforces automatic reconciliation from an authoritative source.

Encryption keys shall be securely stored outside of the systems on which they are used.

Secure Development (if applicable)

The system shall be developed following the 'data protection by design and by default' principle. Appropriate technical and organizational measures shall be in place to implement the data protection principles and safeguard individual rights. Data protection shall be integrated in processing activities and operational practices, from the design stage throughout the solutions lifecycle.

A documented policy and/or process that indicates the use of a known secure development methodology throughout the entire life cycle of the software (SDLC).

The security assurance code must be done using the international and best practices in security, such as, OWASP, PCI, others. Notification and confirmation with the InfoSec team shall be established when a QA security assurance testing is performed.

The system shall be engineered following the 'security by design' principles.		
Development and tests of the system will be done with fictitious or pseudonymized information.		
Access to program source code and associated items - such as designs, specifications, testing and validation plans - shall be strictly controlled; to prevent the introduction of unauthorized functionality.		
The system shall display generic error messages that do not disclose detailed information such as process logs, account or system information.		
Executable code will not be implemented on an operational system until evidence of conforming to the testing criteria (user approval, QA, or the equivalent) is acquired and the associated program source libraries have been updated.		
Security Operations <i>Based on UNICEF's security program requirements, all systems shall follow ICTD and the SOC procedures and provisions to maintain a secured network of interconnected devices internally and externally. The following system component should be considered:</i> • For frontend systems • In between systems / containers • Databases <i>UNICEF / third party hosted solutions shall incorporate appropriate controls to meet this requirement, failure to do so will result in increased risks to operational environment.</i>		
The system shall be hardened, which means that: • only the services and network ports necessary for efficient operation are up and running • all application code is patched and kept up to date and • limiting the accounts and removing, changing, or disabling default accounts and passwords Note: In order to ensure proper risk driven methodology is followed, patches shall fall into one of the following categories, which are classified by the application / system Vendor.; critical, noncritical.		

Servers and applications shall be configured to run with the minimum system authorizations necessary. The service provider shall ensure the implementation of the appropriate technical and organizational measures.		
The production environment shall be separated from the test and development environments; preferably on logically and physically different systems.		
Development and test environment shall have the same patch level as the production environment.		
The production environment shall not have any development tools.		
Configuration/Application source code / customized work, shall be protected from unauthorized access / modification and reside in non-production environment with proper back-up / resiliency policy.		
The system shall have malicious code protection measures. Logs generated by malicious code protection measures shall be monitored.		
Vulnerability Management <i>Assessment of, or validation of connectivity controls between UNICEF and any third party / service provider are required before any physical or logical connection can be made. The following system components shall be considered:</i> • For frontend systems • In between systems / containers <i>Patch management of vulnerable system shall be consistent with UNICEF ICTD patch management process.</i> <i>The service provider is required to report on the results of the security scans and the remediations taken. Any vulnerability identified that affects the scope shall be reported to the UNICEF focal point.</i>		
The service provider is required to run security tests. Test will run prior to the launch of the system and periodically afterwards; with a minimum frequency of once a year.		
The service provider is required to report on the results of the security scans and the remediations taken. These reports will be sent to UNICEF's Chief of IT Security or the relevant focal point(s).		
Critical security patches shall be applied following established testing / change management processes.		
Change Management		

<i>Changes to the system and/or application post-baseline will be documented (i.e. ver. / build number, etc.), along with description via a formal UNICEF change management process for the following system components:</i> • <i>For frontend systems</i> • <i>In between systems / containers</i> • <i>Databases</i>		
Any changes to UNICEF system(s) or software shall be agreed upon between ICT and the business division / office owner of the affected system and third party.		
Changes to system and/or application post baseline will be documented (version / build number), along with description via a formal change management process. The service provider shall report the following information about patches, at a minimum: type, version, reason, post test results after implementation. Patches that fail testing will also be recorded and documented.		
The updating of the operational software, applications and program libraries will only be performed by trained and qualified administrators upon appropriate management authorization.		
Log Management and Monitoring <i>Logs/events will be generated in a format that can be easily parsed and used as an input for logging process management for the following system components:</i> • <i>For frontend systems</i> • <i>In between systems / containers</i> • <i>Databases</i> <i>Event filtering will be enabled on logs for analysis to fine tune the results and consolidate logs into single entry containing count of number of occurrences of events to save space.</i> <i>Analysis will be performed on communication patterns to generate profiles for information systems for detecting unusual activities in the network.</i>		
The system shall generate and process auditing tracks covering all actions taken on personal data, including data access only.		
Authentication validation activities and all changes in authorization shall be logged and securely stored, with limited access.		
Access to content, key information and or any modifications to operational program libraries shall be logged and restricted.		

Logs and events will be generated in a format that can be easily parsed and used as an input for logging process management.		
Integrity log checking shall be performed to ensure consistency.		
The system, application, as well as underlying services and or networks, shall be monitored and activities logged.		
Security Incident Management <i>Reporting of security incidents shall be in coordination with the SOC whether internally or externally.</i> <i>A security breach, shall be viewed as:</i> • a failure in security controls which leads to the accidental, unlawful, or unauthorized access, destruction, loss or alteration of data / information that processed / stored on system • a failure in security controls which leads to the accidental, unlawful, or unauthorized access to ICT resources, such as - but not limited to - computing resources (processing and or storage / services) and communication resources (infrastructure).		
Security breaches, shall immediately be communicated to UNICEF's Point of Contact.		
A security incident notification and escalation procedure shall be formally documented and contractually enforced between the service provider, and UNICEF's Security Operations Centre.		

Table B – Platform and Infrastructure Requirements

Requirement	Complied? (Yes / No)	Proposer's Proposal Page/Section Reference for explanation and supporting documents
The Vendor shall integrate their solution with the established authentication and identity framework; implementation / adoption of Single Sign-On when available.		
Documented firewall considerations and required configurations		
Vendor to provide a clear, publicly stated stance on privacy protection and data use, re: Privacy Policy		
Documented high-availability and disaster recovery capabilities and procedures		
Adherence to General Data Protection Regulation (GDPR) or Department of Defense (DoD) 5220.22-M or National Institute of Standards and Technology (NIST) Special		

Publication (SP) 800-88 for data sanitization on retirement of storage devices		
Publish/provide locations of data centers storing data		
Documentation of data retentions procedures and backup facilities		
Business Continuity Plan including backup and restore procedures		
Describe in detail all the annual support and maintenance schemes that they provide. This will include the internal procedures and processes for resolution of problems and strategies for service improvements etc.		
Clearly define the procedure to handle escalation issues, bugs, and service packs.		

Table C – Privacy and Data Protection Requirements

The following requirements are for implementations that will process personal data which is defined as any information relating to an identified or identifiable individual.

An identifiable individual is one who can be identified, directly or indirectly, in particular by reference to:

- an identifier such as a name, an identification number, audiovisual materials, location data, an online identifier,
- one or more factors specific to the physical, physiological, genetic, mental, economic, cultural, or social identity of the individual, or
- assessments of the status and/or specific needs, such as in the context of assistance programmes.

The definition of what constitutes personal data is contextual and expanding particularly due to enhancements in technology and methods for identifying individuals.

Requirement	Complied? (Yes / No)	Proposer's Proposal Page/Section Reference for explanation and supporting documents
Privacy Governance and Management		
Proposer has implemented a comprehensive privacy program to comply with applicable data protection laws, regulations and international best practices of personal data protection, including a personal data breach procedure. Kindly identify which framework was used for developing such program: • GDPR • Generally Accepted Privacy Principles (GAPP)		

• Health Insurance Portability and Accountability Act (HIPAA) • UN Privacy Principles • Other (please clearly identify)		
Proposer oversees and enforces compliance with the privacy programme. Kindly describe how such activities are carried out and specify the frequency and audit activities performed, i.e., whether through internal and/or external auditors		
Proposer agrees not to use any data (even if anonymized) processed for any other purpose than for the purposes of the Proposal (e.g., AI/machine learning, deriving other data, etc.)		
The solution complies with best industry standards of “privacy by design and default”, meaning any processing of personal data from users of the platform should follow international best practice of data protection and configured in a least privacy intrusive manner.		
Describe the implementation of an easily visible and understandable appropriate “opt-out” mechanism when collecting consent.		

[Appendix 3 - CLASSI III UNICEF Security Requirements](#)

[Appendix 4 - QA test plan](#)

[Appendix 5 - Sample functional and non-functional test cases](#)